

Ankita Das

Curriculum Vitae

SRM University AP, Amaravati
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Research interest

My research interest widely lies in algebraic number theory. Currently I am interested in Jacobian varieties of super elliptic curves. Also I like to explore different aspects of algebraic geometry and the interplay between geometry and number theory.

Academic information

- 2024-Present **PhD in Mathematics**, SRM University, AP, and Indian Institute of Science Education and Research(IISER),Thiruvananthapuram
- 2020-2022 **Master of Science (M.Sc.) in Mathematics**, Jadavpur University
- 2017-2020 **Bachelor of Science (B.Sc.) in Mathematics**, Jadavpur University
- 2017 **Higher Secondary Examination (10+2 level)**, West Bengal Board, Sunity Academy
Percentage – 93.6
- 2015 **Secondary Examination (10 level)**, West Bengal Board, Sunity Academy
Percentage – 95.6

Papers published

- 2026 **Generalized Fruit Diophantine Equation And Super Elliptic Curves**, Published in *Monatshefte Fur Mathematik*, <https://doi.org/10.1007/s00605-026-02201-6>, with Prof. Kalyan Banerjee and Prof. Kalyan Chakraborty

Manuscripts

- 2026 **Rational Torsion and Ramification Constraints on Odd Superelliptic Jacobians**

Projects

- Ongoing work in PhD **Solvability problems in *Diophantine equations* via the arithmetic of elliptic and superelliptic curves**, with Prof. Kalyan Banerjee and Prof. Kalyan Chakraborty
- Master's Project **A Survey on Clean Ring and Its Structural Behaviour**, link: https://drive.google.com/file/d/1QEcnNCNo7x_4B1HTCkSsJS3VSQXVSbH/view?usp=drivesdk, under supervision of Prof Sukhendu Kar

Some slides of my talks

- **On Jacobian on a Family of Superelliptic Curves**, link: https://drive.google.com/file/d/1lvwSfr17jE-k-ZNXUjwckJ3xArooIjCm/view?usp=drive_link
- **Solvability problems in *Diophantine equations* via the arithmetic of elliptic and superelliptic curves**, link: <https://drive.google.com/file/d/1nq56pQEbzHCpuFzFdTPHH0ibFv4wzvHf/view?usp=drivesdk>
- **From Fruits to Class Groups**, link: https://drive.google.com/file/d/16ayXCVwTSfUipIeFV88a9r4S6PWTUVU_Q/view?usp=drivesdk
- **Factorization of Sine and Cosine functions and Proof of Walli's Formula**, link: https://drive.google.com/file/d/1EK-sX0J0REoSGXfnHoqfGao__CXsdR4g/view?usp=drive_link
- **Zariski-Nagata Theorem: A Few Consequences**, link: https://drive.google.com/file/d/1mKtPXQp_n2EJIIdaNx6fh-ftqd4r_NaHI/view?usp=sharing

Courses Taken in PhD Coursework

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|-----------------------------|-----------------------------------|
| ○ Fourier Analysis | ○ Category Theory |
| ○ Advanced Complex Analysis | ○ Representation Theory |
| ○ Advanced Measure Theory | ○ Infinite Galois Extension |
| ○ Commutative Algebra | ○ Cohomological Algebra |
| ○ Differential Geometry | ○ Research Methodology |
| ○ Manifold | ○ Research Publication and Ethics |
| ○ Algebraic Number Theory | ○ Research Seminar |
| ○ Algebraic Geometry | |

Courses Taken in Graduate Studies

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|---|---------------------------------------|
| ○ Abstract Algebra | forms and Calculus of Variations |
| ○ Real Analysis | ○ Algebraic and Differential Topology |
| ○ Topology | ○ Advanced Rings and Modules-I |
| ○ Linear Algebra | ○ Theory of Semigroups-I |
| ○ General Mechanics | ○ Optimization and Control Theory |
| ○ Differential Geometry, Probability and Stochastic Process | ○ Advanced Functional Analysis |
| ○ Linear Algebra and Module Theory | ○ Advanced Rings and Modules-II |
| ○ Complex Analysis and PDE | ○ Theory of Semigroups-II |
| ○ Functional Analysis | ○ Topics in Analysis |
| ○ Ordinary Differential Equations and Special functions | ○ Topics in Linear Algebra |
| ○ Advanced Topology | ○ Topics in Manifold |
| ○ Dynamical System | ○ Commutative Algebra |
| ○ Integral Equations, Integral Trans- | ○ Topics in Algebra |
| | ○ Differential Geometry |

Courses Taken in Undergraduate Studies

- Algebra, Geometry and Calculus
- Abstract Algebra
- Statistics
- Real Analysis
- Ordinary Differential Equations
- Computer Science
- Linear Algebra
- Numerical Methods
- Partial Differential Equations
- Metric Space and Complex Analysis
- Linear Programming and Game Theory
- Probability
- Mechanics
- Number Theory
- Combinatorics and Graph Theory

Exams Qualified

- 2023 **Qualified Joint CSIR-UGC NET**
- 2023 **Qualified West Bengal State Level Eligibility Test, (WBSET)**
- 2021 **Qualified IIT JAM examination**
- 2017 **Obtained 94% marks in Mathematics, 90% marks in Physics in 10+2 standard.**
- 2015 **Obtained 100% marks in both Mathematics and Physical Science in 10 standard.**

Awards and Scholarships

- 2025 **President Scholastic Award, as "*Maintenance Allowance*" for securing external government funding, by SRM University, AP**
- 2023-Present **Joint CSIR-UGC JRF Fellowship, for qualifying all India *Joint CSIR-UGC NET*, by UGC**
- 2022-2021 **Indira Gandhi Single Girl Child Scholarship, by UGC**
- 2022-2017 **Bigyani Kanya Medha Britti (BKMB) scholarship, for securing position in *top 50 female candidates* in a state level exam organized by JBNSTS**
- 2017 **DST-INSPIRE Scholarship, for securing position in *top 1 percent candidates* in 10+2 level state board examination, by Scholarship for Higher Education (SHE)**

Work Experience

- Odd Semester 2026-2027 **TA for grading, CSE 203 (*Discrete Mathematics*), SRM University, AP**
- Odd Semester 2025-2026 **TA for grading, CSE 203 (*Discrete Mathematics*), SRM University, AP**
- 2022 **Gramin Dak Sevak, Assistant Branch Post Manager, Indian Sub Post Office, Nilkuthi, Cooch Behar**

Technical Skills

Programming Languages **LaTeX, C, C++, Python**

Conferences and Symposiums

- July, 2026 **41st Annual Conference of Ramanujan Mathematical Society (RMS)**, *India*
- December, 2025 **Indian Women and Mathematics (IWM), Research in Groups programme**, *India*
- December, 2025 **Sakura Science Exchange Program**, *Japan*
- October, 2025 **International Conference on Class Groups Of Number Fields And Related Topics (ICCGNFRT)-2025**, *SRM University, AP, Andhra Pradesh, India*
- January, 2025 **Frontier Symposium in Mathematics (FSM)-2025**, *IISER Thiruvananthapuram, Kerala, India*
- October, 2023 **International Conference on Class Groups Of Number Fields And Related Topics (ICCGNFRT)-2025**, *Kerala School of Mathematics (KSoM), Kerala, India*

Extra Curricular Activities

- I was trained in **Hindustani Classical Music** and **Rabindrasangeet** for several years. I like to listen various forms of semi-classical and light music. I enjoy different classical instruments. I won several district and state level music competitions.
- I was trained in **recitation** from Bangiya Sangeet Parishad and also completed my **Visharad** in it. I won various state and national level recitation competitions. I was in **AIR** as child recitation artist.
- I won a national essay writing competition on "**Eradication of Poverty**" organised by **TATA building India** and felicitated by **President of India** in 2014.
- Writing poetry, stories and reading books give me immense pleasure.
- I always enjoy to take part in instantaneous debate and won many district level competitions.

Languages

English, Hindi, Bengali